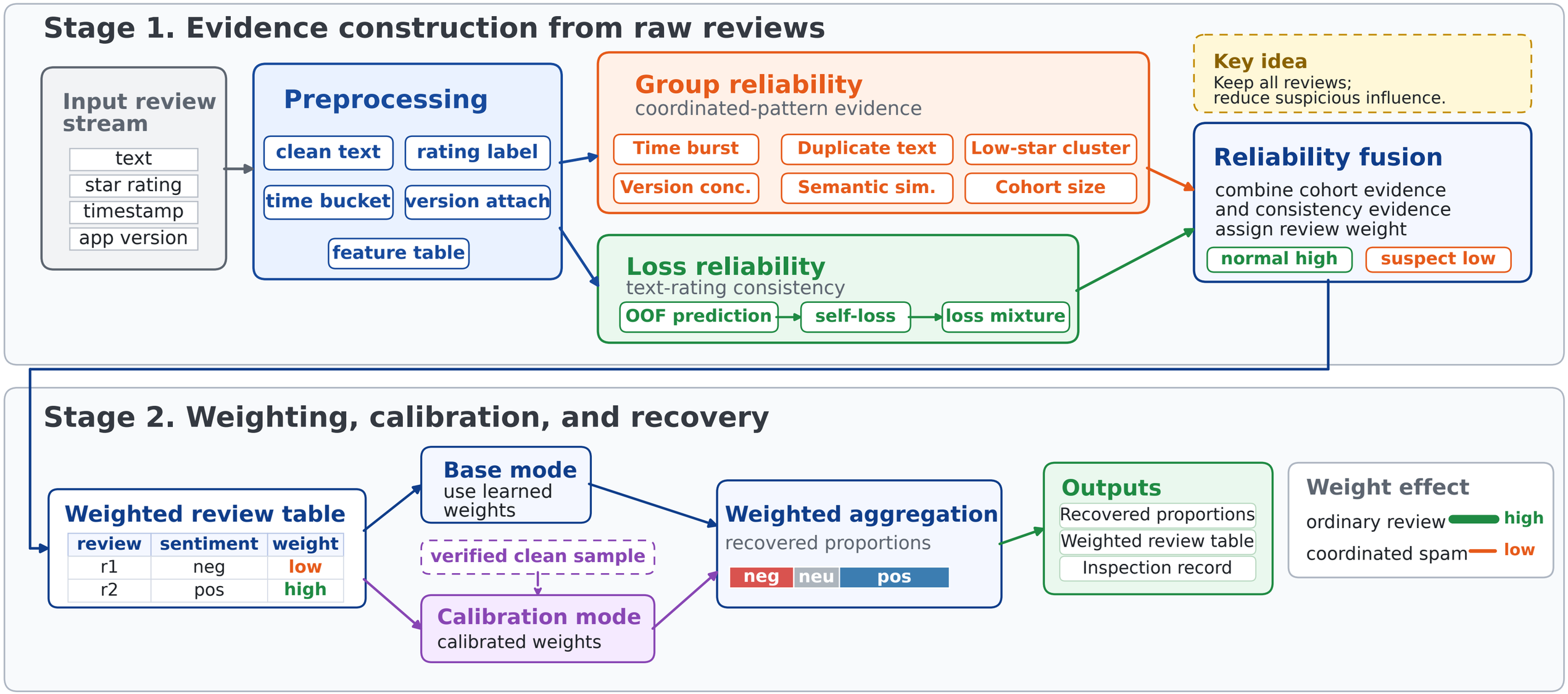
Stage 1. Evidence construction from raw reviews
Input review stream
text
star rating
timestamp
app version
Preprocessing
clean text
rating label
time bucket
version attach
feature table
Group reliability
coordinated-pattern evidence
Time burst
Duplicate text
Low-star cluster
Version conc.
Semantic sim.
Cohort size
Loss reliability
text-rating consistency
OOF prediction
self-loss
loss mixture
Key idea
Keep all reviews;
reduce suspicious influence.
Reliability fusion
combine cohort evidence
and consistency evidence
assign review weight
normal high
suspect low
Stage 2. Weighting, calibration, and recovery
Weighted review table
review
sentiment
weight
r1
neg
low
r2
pos
high
Base mode
use learned weights
verified clean sample
Calibration mode
calibrated weights
Weighted aggregation
recovered proportions
neg
neu
pos
Outputs
Recovered proportions
Weighted review table
Inspection record
Weight effect
ordinary review
high
coordinated spam
low
Legend:
blue = preprocessing
purple = optional calibration
orange = cohort evidence
solid arrow = main flow
green = consistency evidence
dashed arrow = optional input